

Assignment 3 – Data preparation

Description

We will prepare data to be used in both MySQL and MongoDB. You must use the data dumps provided in myCourses for both databases.

Your tasks

1.- Provide programs that build single SQL and MongoDB queries to retrieve the following:

- 1.1.- For a given range of dates, the histogram of reviews by business category.
- 1.2.- For a given range of dates, 2D histogram of reviews by business category and year.
- 1.3.- For a given business category, p-th quantile of the review ratings.
- 1.4.- For a given business category, mode of the price range.

Check the grading software and software templates for more info.

2.- Provide programs that build single SQL and MongoDB queries to scale the following:

- 2.1.- For a range of “yelping since” dates, min-max scaling of funny votes.
- 2.2.- For a range of review dates, mean scaling of funny votes given to reviews.
- 2.3.- For a range of “yelping since” dates, z-score scaling (either standard or squared) of cool compliments.
- 2.4.- For a business category, robust scaling of latitudes.

Check the grading software and software templates for more info.

3.- Provide programs that build single SQL and MongoDB queries to compute the following bins:

- 3.1.- For a range of “yelping since” dates, equal-width bins of funny votes using k bins.
- 3.2.- For a range of “yelping since” dates, equal-depth bins of funny votes using k bins.

Check the grading software and software templates for more info.

Submission instructions

- Use the software template(s) and grading software provided in myCourses.
- Submit a single ZIP file to myCourses that must be named as your RIT user, e.g., `errves.zip`. Do not include ‘@rit.edu.’ The file must contain folders named as the folders provided in the software template(s) containing your queries, Gradle or Poetry project(s).
- Everything will be graded on a Linux machine, so you must always use the exact names provided in this document, software template(s) and grading software.

Grading rubric

- All test cases in the grading software must pass with some restrictions like main memory and total running time.